

NLP - 2026, Quiz - 2

1. How do we evaluate ASR and TTS models? Name them. [2]

ASR: _____ WER (Word Error Rate) _____

TTS: _____ MOS (Mean Opinion Score) _____

2. Distinguish the following based on their density, semantic, and dimensionality in 2-3 words only.

a. One-hot, Co-occurrence, and Word embeddings.

[3]

	Density	Semantic	Dimensionality
One-hot vector	Sparse	No semantic	V for one word OR V x V for vocabulary
Co-occurrence vector	Partially sparse/dense	Low semantic	V for one word OR V x V for vocabulary
Word embeddings	Dense	High Semantic	d for one word OR V x d for vocabulary

3. Define your NLP course project considering the following details:

- a. Project Title (1 line) [1]
- b. Describe your chosen problem statement (2-3 lines) [3]
- c. Number of samples you used for testing / evaluation, i.e., samples in the test set (just number)
– We need numbers (approximately within +/- 5% of the test samples). Percentage will not work. [2]
- d. Name of evaluation metric used (Just name) [2]
- e. Which baseline did you use for mid-project submission? Describe it in 1-3 lines. [2]

As per your submitted mid-project report

4. Define the following terms with toy examples: [3]

- a. Zero-shot setting
The model is prompted without providing any example of the task. E.g.,
Predict $f(x)$.
- b. One-shot setting
The model is prompted with one example of the task to learn and respond accordingly. E.g.,
If $f(x_1) \rightarrow y_1$, then predict $f(x_2)$
- c. Few-shot setting
The model is prompted with multiple examples of the task to learn and respond accordingly. E.g.,
If $f(x_1) \rightarrow y_1, f(x_2) \rightarrow y_2, \dots, f(x_n) \rightarrow y_n$ then predict $f(x_m)$

5. We learn LSTM for ELMo, whereas, for training word2vec, we define a context-size c centered around a word w_t and compute

$$\frac{1}{T} \sum_{t=1}^T \sum_{-c \leq j \leq c, j \neq 0} \log p(w_{t+j} | w_t)$$

Which model belongs to contextual word embedding and which one belongs to non-contextual word embeddings? [2]

Contextual: ELMo

Non-Contextual: word2vec